SIT720 11.1HD

Machine Learning Mini Research

Reproduction and Leakage-Aware Extension of
"An efficient stacking-based ensemble technique for early heart attack prediction"

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Unit: SIT720 - Machine Learning

Task: 11.1HD

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Execution verified
The notebook has been executed successfully from start to finish using the exact 1,025-row heart.csv. The dataset checksum matches the archived public copy (MD5 6481ed154d5b8ff83d7c09007aaa6ed0). The run found 723 exact duplicate rows and 302 unique records. All numerical results and figures in this report were generated by the executed notebook. The Panopto video presentation URL and the accessible OneDrive folder containing the complete code, dataset and README files have both been added for OnTrack submission.

Executive Summary

This study reproduces and critically evaluates the stacking-based heart-disease classifier proposed by Bhagat, Sharma and Agarwal [1], then develops a leakage-aware alternative. Using the checksum-verified 1,025-row dataset and the documented reproduction assumptions, the reproduced stacking model achieved 98.54% accuracy, precision 1.0000, recall 0.9709, F1 0.9852, and AUC 1.0000. The published stacking values are 98.53% accuracy, precision 1.0000, recall 0.9727, F1 0.9861, and AUC 0.9880 [1]. Thus, the reproduced stacking accuracy is almost identical to the paper, although several individual reproduced models are much stronger than the published individual-model results.

The reproduction is deliberately transparent about missing implementation details. The paper does not report the random seed, exact classifier hyperparameters, exact scaling method, or the precise level-2 learner configuration. The executed notebook therefore records explicit assumptions rather than presenting them as facts from the article. A fixed random_state=42 is used for reproducibility, the inferred 80/20 split is retained for the paper-style reproduction, and all assumptions are listed in the protocol table below.

The main validity concern is duplicated observations. The executed data audit found 723 exact duplicate rows, leaving only 302 unique records, consistent with independent discussion of this compiled heart dataset [3]. Random row-level splitting can therefore place identical clinical profiles in both training and test sets. This concern is visible empirically: across 30 random seeds, the reproduced stack achieved 100% test accuracy in 25 splits, with a minimum accuracy of 97.07%. When evaluation is repeated only on unique records, the paper-style stack falls to 80.33% accuracy and AUC 0.8734. This does not prove that duplication is the only cause, but it demonstrates that the headline result is highly sensitive to the evaluation protocol.

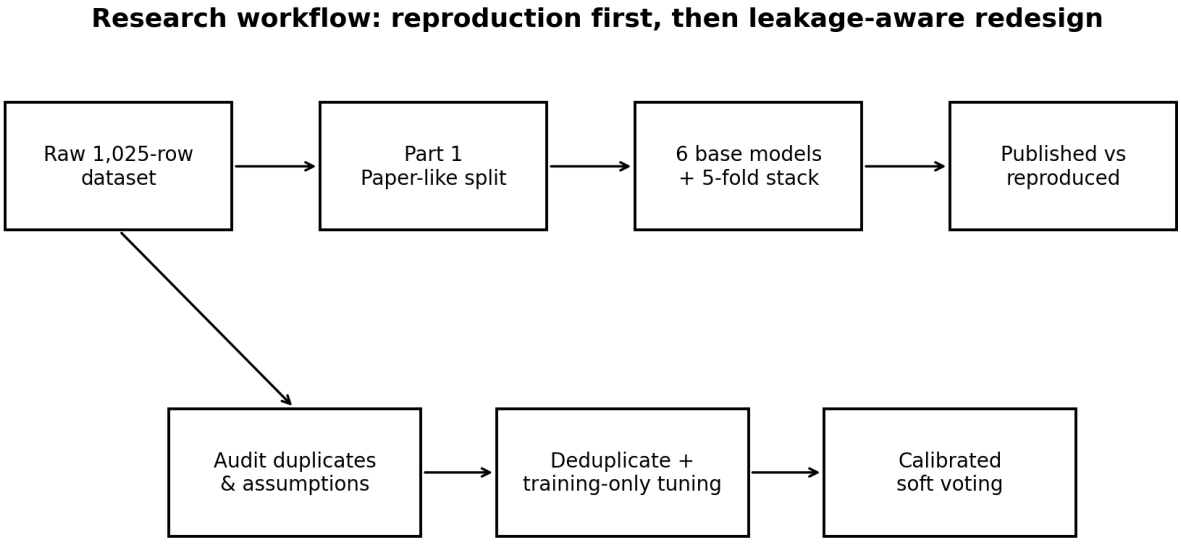


Figure 1. Overall workflow used in this research: faithful reproduction followed by a leakage-aware redesign.

1. Research Problem, Motivation and Research Gap

Cardiovascular disease prediction is a binary classification problem in which routinely collected clinical variables are used to estimate whether a patient has heart disease. The selected article motivates the problem by the global burden of heart disease and the value of early risk identification. It argues that machine learning can support clinical decision making by learning multivariable relationships that are difficult to capture with simple manual rules [1].

The article positions ensemble learning as its main methodological contribution. Six individual classifiers are trained after preprocessing, and their predictions are then combined using stacking. The intended benefit is that a meta-classifier can exploit complementary strengths of different learners and improve prediction accuracy [1].

The paper reviews prior work using feature selection, random forests, support-vector machines, PCA, majority voting, and other ensemble approaches. Reported accuracies in the cited literature vary substantially across datasets and protocols. This variation is important because performance values from different datasets or validation designs are not directly comparable. The present reproduction therefore focuses on the exact 1,025-row dataset and the paper's own model family rather than treating cross-paper accuracy values as equivalent evidence.

The principal research gap addressed here is not simply whether a different classifier can produce a higher score. The gap is whether the reported near-perfect result remains credible under a leakage-aware and fully specified evaluation protocol. Data leakage is known to produce overly optimistic machine-learning results and to undermine reproducibility across scientific fields [4].

2. Selected Paper and State-of-the-Art Context

Bhagat et al. [1] combine six widely used supervised classifiers—LR, DT, RF, XGB, NB and KNN—with stacking. Their literature review describes several contemporary directions in heart-disease prediction: feature selection with LASSO or Relief, PCA-based dimensionality reduction, SVM-based diagnostic systems, random-forest hybrids, and ensemble strategies such as majority voting and stacked models. The paper's stated objective is to produce a highly accurate heart-disease predictor and to show that stacking improves upon individual models.

The paper's own reported results are unusually high for the proposed stack. Table 11 reports 98.53% accuracy and 0.988 AUC for the stacking model, while RF and DT are reported at 92.68% accuracy and XGB at 90.73% [1]. However, the introduction contains inconsistent values—93.17% for XGB and 96.58% for stacking—whereas the abstract, Results section, Table 11 and conclusion state 90.73% and 98.53%, respectively. The later correction notice addresses missing text or wording in Tables 2, 3, 6 and 7, not Table 11 or these result discrepancies [7].

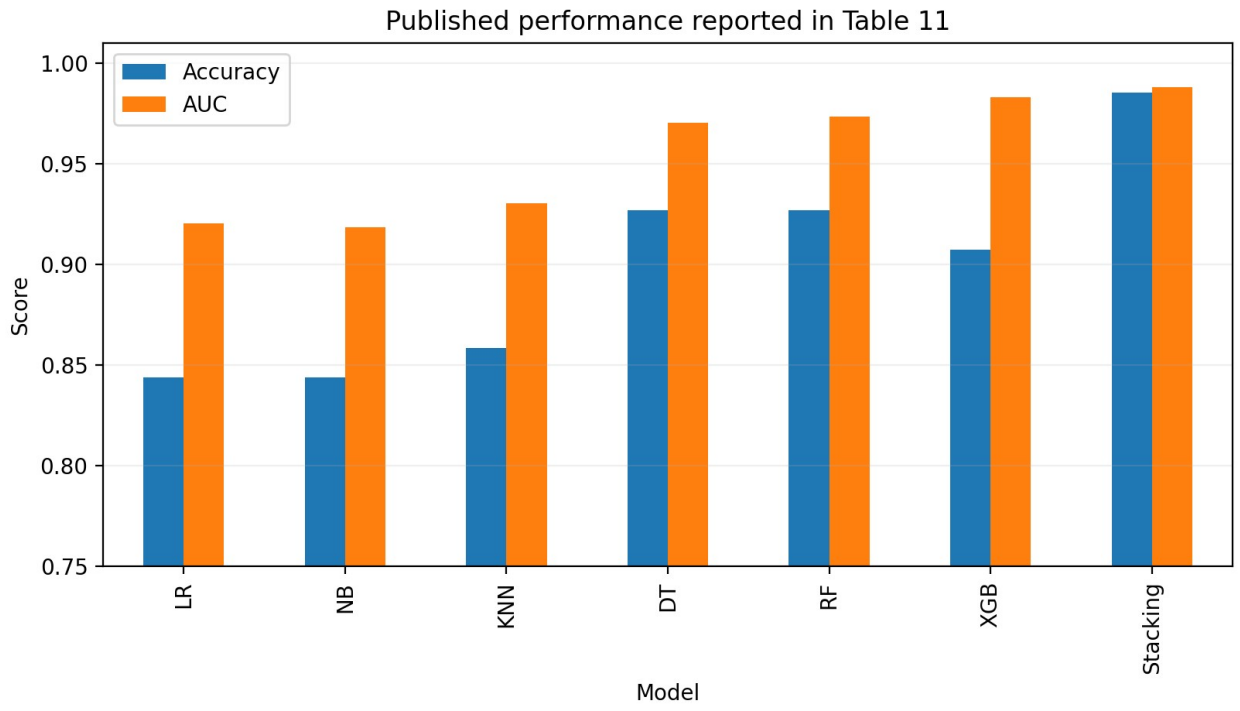


Figure 2. Core published Accuracy and AUC values transcribed from Table 11 of the selected article [1].

3. Dataset and Feature Set

The paper points to the public Kaggle dataset johnsmith88/heart-disease-dataset and describes it as a combination of Cleveland, Hungary, Switzerland and Long Beach V sources. It states that the source contains 76 characteristics but that 14 columns, including the target, are used in the experiments [1]. The UCI Heart Disease repository similarly describes four contributing databases and notes that published experiments commonly use a 14-attribute subset [2].

Feature	Meaning	Paper range/encoding
age	Age in years	29-77
sex	0=female, 1=male	0-1
cp	Chest pain type	0-3
trestbps	Resting blood pressure (mmHg)	94-200
chol	Serum cholesterol (mg/dl)	126-564
fbs	Fasting blood sugar >120 mg/dl	0-1
restecg	Resting ECG result	0-2
thalach	Maximum heart rate achieved	71-202
exang	Exercise-induced angina	0-1
oldpeak	Exercise-induced ST depression	0-6.2
slope	Slope of peak exercise ST segment	0-2
ca	Number of major vessels	paper: 0-3
thal	Thalassemia category	paper: 1-3
target	Heart disease class	0=no, 1=disease

A reproducibility audit should also check observed ranges rather than assuming the paper's table is exact. Public versions of the 1,025-row CSV can contain ca=4 and thal=0, which are outside the ranges stated in the paper. The notebook therefore records the actual minimum, maximum and number of unique values for every column.

For execution, the local heart.csv was verified against the archived public copy in [9] using MD5 6481ed154d5b8ff83d7c09007aaa6ed0. This provides a reproducible identity check for the exact 1,025-row file used in all reported experiments.

The executed dataset audit confirms the duplication issue directly: 723 rows are exact duplicates and only 302 unique rows remain. Because duplicates can cross an ordinary random train/test boundary, a model may be tested on records that are identical to records seen during training. The proposed experiment therefore removes exact duplicates before any split and keeps the final test set untouched during tuning.

4. Original Machine-Learning Methods and Experimental Protocol

The original workflow is: preprocess missing values, normalise numerical attributes, encode categorical variables, split the data into training and test sets, train six classifiers, and then combine base predictions through stacking [1]. The paper states that a basic linear-regression imputation method is used for missing values and explains stacking conceptually using fold-based out-of-sample predictions. In the Results section it describes the proposed method as a 5-fold stacking ensemble.

The article does not provide enough implementation detail for exact bit-for-bit reproduction. In particular, it does not identify the random seed, explicit train/test ratio, exact normalisation method, exact hyperparameters for each model, or exact level-2 meta-classifier. The task brief explicitly permits reasonable assumptions when a paper is unclear, provided those assumptions are justified. Accordingly, the reproduction assumptions are fixed before evaluation rather than adjusted after seeing test results.

Component	Reproduction choice	Justification
Train/test split	80/20, unstratified	Published decimal scores map exactly to 205 test cases; reported precision/recall are consistent with 110 positives and 95 negatives.
Random seed	42	The paper does not report one; a fixed seed is required for reproducibility.
Continuous scaling	StandardScaler on age, trestbps, chol, thalach, oldpeak	The paper says numerical attributes are normalised but does not name a scaler.
Categorical encoding	Use the dataset's existing integer encoding	The downloaded file is already numerically encoded.
Missing values	Regression imputation if present; otherwise no-op	Matches the paper's described imputation without inventing missing data.
Base-model parameters	Library defaults where unspecified	Avoids reverse-engineering parameters from test performance.
Stacking	All six base models, 5-fold CV, logistic-regression meta-model	Five folds are stated; the exact meta-model is omitted, while LR is described as a valid meta-model in the paper.

5. Published Results and Critical Reproducibility Audit

Model	Accuracy	Precision	Recall	F1	AUC
LR	0.8439	0.8196	0.9090	0.8620	0.9204
NB	0.8439	0.8250	0.9000	0.8608	0.9185
KNN	0.8585	0.8648	0.8727	0.8687	0.9304
DT	0.9268	0.9702	0.8909	0.9289	0.9702
RF	0.9268	0.9130	0.9545	0.9333	0.9734
XGB	0.9073	0.9174	0.9090	0.9132	0.9830
Stacking	0.9853	1.0000	0.9727	0.9861	0.9880

A useful reproducibility check is to ask whether the published decimals correspond to an integer confusion matrix. For the six individual models, the Accuracy/Precision/Recall triplets are exactly consistent with a 205-case holdout containing 110 positives and 95 negatives. For example, LR corresponds to TP=100, FN=10, FP=22 and TN=73; RF corresponds to TP=105, FN=5, FP=10 and TN=85. This strongly supports an 80/20 split even though the paper does not state the ratio clearly.

Model	TP/FN/FP/TN*	Paper specificity	Derived specificity	Paper MCC	Derived MCC
LR	100/10/22/73	0.8620	0.7684	0.8196	0.6883
NB	99/11/21/74	0.8608	0.7789	0.8250	0.6872
KNN	96/14/15/80	0.8687	0.8421	0.8648	0.7154
DT	98/12/3/92	0.9289	0.9684	0.9702	0.8571
RF	105/5/10/85	0.9333	0.8947	0.9130	0.8534
XGB	100/10/9/86	0.9132	0.9053	0.9174	0.8138
Stacking	107/3/0/95	0.9861	1.0000	1.0000	0.9711

*Confusion counts are reconstructed from the paper's Accuracy/Precision/Recall values on a 205-case test set. The Stacking row uses the 110-positive/95-negative composition implied by the base-model rows.

The derived specificity and MCC values do not match the corresponding columns of Table 11. In several rows, the paper's "Sensitivity" repeats Accuracy, "Specificity" repeats F1, and "MCC" repeats Precision. Because Table 10 gives the standard formulas, this pattern is more consistent with a reporting/transcription problem than with alternative metric definitions. The reproduction therefore computes every metric directly from predictions and a confusion matrix.

6. Reproduction Study

The notebook was executed successfully using the six classifiers and a 5-fold stacking model under the assumptions in Section 4. The reproduced stacking model produced TN=102, FP=0, FN=3, and TP=100. Its accuracy was 98.54%, F1 was 0.9852, and AUC was 1.0000. The notebook also exports the full model table, ROC curves, stacking confusion matrix, seed-sensitivity analysis, and machine-readable result files.

Model	Accuracy	Precision	Recall	F1	AUC
LR	0.7951	0.7563	0.8738	0.8108	0.8769
NB	0.8000	0.7541	0.8932	0.8178	0.8706
KNN	0.8049	0.7788	0.8544	0.8148	0.9421
DT	0.9854	1.0000	0.9709	0.9852	0.9854
RF	0.9854	1.0000	0.9709	0.9852	1.0000
XGB	0.9854	1.0000	0.9709	0.9852	0.9894
Stacking	0.9854	1.0000	0.9709	0.9852	1.0000

Table: Executed reproduction results (random_state=42). The stacking model reaches 98.54% accuracy; its test confusion matrix is TN=102, FP=0, FN=3, TP=100.

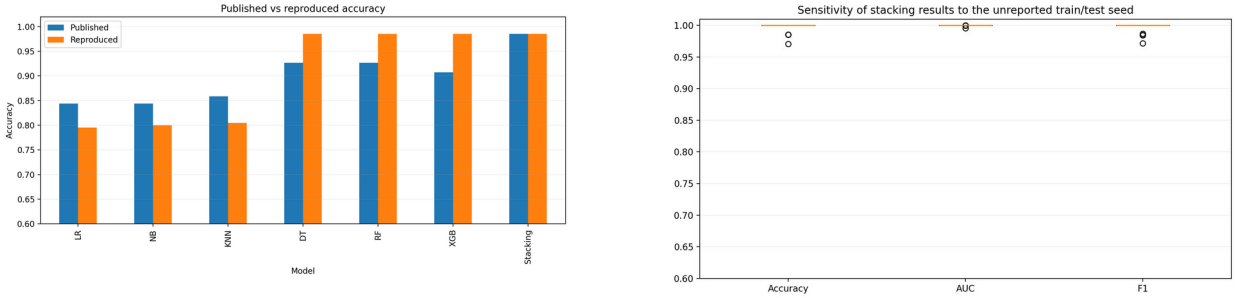


Figure 3. Published-versus-reproduced accuracy (left) and stacking accuracy sensitivity across 30 random seeds (right).

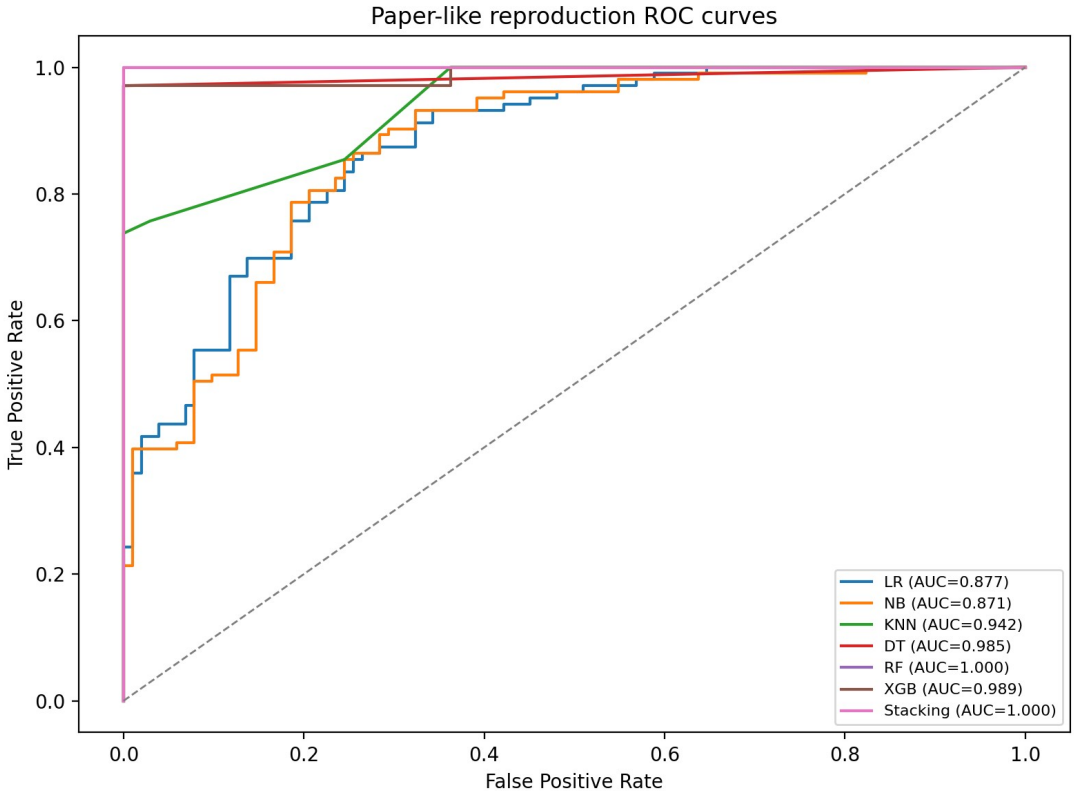


Figure 4. ROC curves from the executed paper-style reproduction on the duplicate-rich random split.

The reproduced stacking accuracy differs from the published value by only 0.007 percentage points. However, the individual models do not reproduce the published ranking: DT, RF and XGB each reach 98.54% accuracy in this particular split, compared with 92.68%, 92.68% and 90.73% respectively in Table 11 [1]. Four plausible causes are supported by the audit: the undocumented random split, unspecified hyperparameters, ambiguities in preprocessing, and extensive exact duplication. The 30-seed sensitivity experiment reinforces that a duplicate-rich random split frequently yields near-perfect results.

7. Critical Analysis of the Reproduced Work

Strengths of the paper include the use of several heterogeneous classifiers, the attempt to compare more than accuracy alone, and the use of an ensemble rather than relying on one model family. The feature

set is clinically interpretable and small enough for transparent experimentation. The reported stack also demonstrates the intuitive value of combining different decision boundaries.

However, the reproducibility limitations are material. The missing seed and parameters prevent an exact reconstruction; inconsistent result values appear across sections; and Table 11 contains metric columns that are not numerically consistent with its own Accuracy/Precision/Recall values. Most importantly, the public 1,025-row dataset is duplicate-heavy. Leakage literature shows that train-test contamination can make scientific ML performance substantially overoptimistic [4].

The duplicate issue changes the interpretation of the 98.53% accuracy. It should not be interpreted as evidence that the system would achieve 98.53% accuracy on genuinely new patients. A defensible conclusion is narrower: the paper reports very high performance under its stated row-level experiment, but the absence of a leakage-aware protocol means external generalisation remains uncertain.

Model-selection methodology also matters. If hyperparameters or model choices are adjusted using the same folds later used to estimate performance, error estimates can be optimistically biased. Varma and Simon [5] show why model selection should be nested inside an outer evaluation process. The proposed method therefore isolates a final holdout and performs tuning only on training data.

8. Proposed ML Solution: Leakage-Aware Calibrated Soft Voting (LACSV)

The proposed system is designed to address methodological rather than cosmetic limitations. It differs from the paper in data preparation, validation design, ensemble structure and uncertainty reporting.

1. Exact duplicate removal before any split. This prevents identical rows from being shared between training and test data.
2. Untouched stratified 20% final test set created from unique records. All model selection is confined to the remaining training set.
3. Training-only repeated stratified 5-fold cross-validation for hyperparameter selection using ROC-AUC.
4. Three complementary learners: regularized Logistic Regression, RBF-SVM, and Random Forest. Their parameters are tuned using only training folds.
5. Cross-validated sigmoid calibration of each learner's predicted probability. Probability calibration is relevant when outputs may support risk-based decisions; prior work shows that some learning algorithms can produce distorted probability estimates and can benefit from calibration [6].
6. Soft voting over calibrated probabilities instead of a level-2 stacking meta-classifier.
7. Fair baseline comparison: the paper-style stack is re-evaluated on the exact same deduplicated train/test split.
8. Bootstrap 95% confidence intervals for Accuracy, F1 and AUC to show uncertainty due to the small leakage-free test set.

This is a substantial methodological contribution because it changes what counts as an independent observation, how model selection is separated from final testing, how base probabilities are combined, and how uncertainty is reported. The objective is not to maximize an inflated score; it is to obtain a more credible estimate of performance on unseen clinical profiles.

9. Proposed Experimental Protocol and Results

Element	Setting

Raw data	Exact paper-linked 1,025-row heart.csv
Cleaning	Remove exact duplicate rows before splitting
Final holdout	20%, stratified, random_state=42, never used for tuning
Training validation	RepeatedStratifiedKFold: 5 folds × 3 repeats
Tuning score	ROC-AUC
LR grid	C ∈ {0.05,0.1,0.5,1,2,10}; class_weight ∈ {None, balanced}
RBF-SVM grid	C ∈ {0.1,0.5,1,2,10}; gamma ∈ {scale,0.01,0.05,0.1}; class_weight
RF grid	n_estimators ∈ {200,500}; max_depth ∈ {None,3,5,8}; min_samples_leaf ∈ {1,2,4}; class_weight
Calibration	Sigmoid, 5-fold CV on training data
Final metrics	Accuracy, Precision, Recall, F1, AUC, Specificity, MCC, Brier score
Uncertainty	2,000 bootstrap resamples for 95% CI on Accuracy, F1, AUC

Model	Accuracy	Precision	Recall	F1	AUC	Brier
Paper-style stack	0.8033	0.8387	0.7879	0.8125	0.8734	0.1487
Proposed LACSV	0.8033	0.8182	0.8182	0.8182	0.8712	0.1447

Table: Leakage-free holdout comparison. Lower Brier score is better for probability accuracy.

Base learner	Best training CV AUC	Selected parameters
LR	0.9090	{"model__C": 0.5, "model__class_weight": null}
SVM	0.9126	{"model__C": 2, "model__class_weight": null, "model__gamma": "scale"}
RF	0.9206	{"model__class_weight": "balanced", "model__max_depth": 5, "model__min_samples_leaf": 4, "model__n_estimators": 500}

Table: Training-only cross-validation used to tune the three LACSV base learners.

Model	Metric	Estimate	Bootstrap 95% CI
Paper-style stack	Accuracy	0.8033	[0.7049, 0.9016]
Paper-style stack	F1	0.8125	[0.6909, 0.9062]
Paper-style stack	AUC	0.8734	[0.7720, 0.9505]
LACSV	Accuracy	0.8033	[0.7049, 0.9016]
LACSV	F1	0.8182	[0.7018, 0.9067]
LACSV	AUC	0.8712	[0.7688, 0.9510]

Table: Bootstrap uncertainty on the leakage-free test set. The intervals overlap strongly.

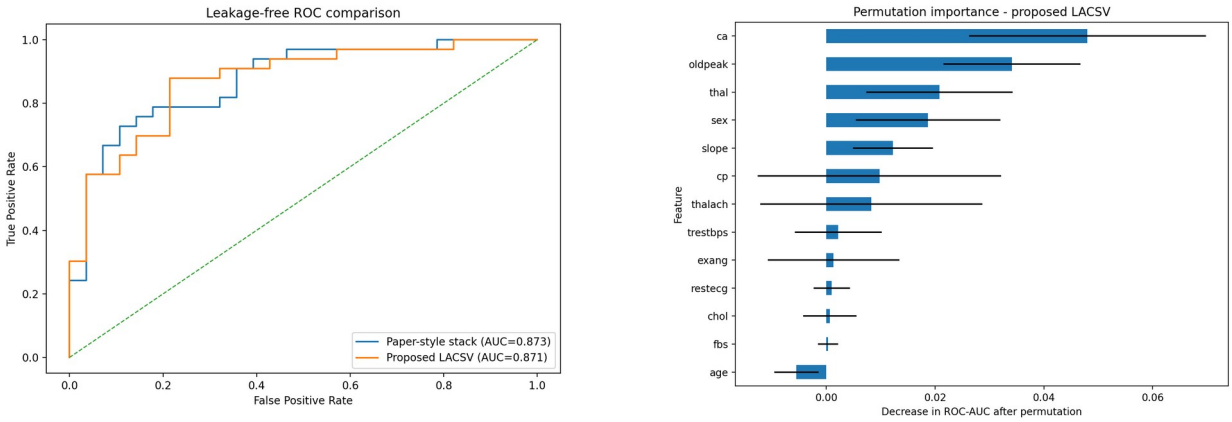


Figure 5. Leakage-free ROC comparison (left) and LACSV permutation importance on the held-out test set (right).

The fair comparison uses the same deduplicated train/test split for both methods. The paper-style stack and LACSV both achieved 80.33% accuracy. LACSV increased recall from 0.7879 to 0.8182 and F1 from 0.8125 to 0.8182, while AUC changed from 0.8734 to 0.8712. Its Brier score was slightly lower (0.1447 versus 0.1487), indicating marginally better probability calibration on this holdout. The bootstrap confidence intervals overlap substantially, so these results do not justify claiming that LACSV is decisively superior; its principal contribution is a more defensible leakage-aware evaluation and calibrated decision-support pipeline.

10. Strengths, Limitations and Practical Implications

The main strength of the proposed design is validity: exact duplicates are removed before evaluation, all hyperparameter selection occurs inside training data, the final test set remains untouched, and probabilities are calibrated before soft voting. On the fair holdout, LACSV achieved precision 0.8182, recall 0.8182, F1 0.8182, AUC 0.8712, and Brier 0.1447. Permutation importance identifies ca, oldpeak, thal, sex and slope among the more influential predictors for this fitted system; these importances are model-specific associations, not causal clinical effects.

The study remains limited by the underlying dataset. After deduplication, only about 302 unique records remain, so statistical uncertainty is substantial and subgroup performance cannot be estimated reliably. The dataset is also an old, compiled benchmark rather than a contemporary prospective clinical cohort. The target is a simplified binary label, and the experiment does not establish clinical utility, causal risk factors, transportability across hospitals, or safety for autonomous diagnosis.

Accordingly, the model should be described as an educational/research classifier or decision-support prototype, not as a clinical diagnostic system. A credible next study would evaluate a larger, independently collected cohort, lock the model before external testing, assess calibration and subgroup performance, and conduct prospective validation.

11. Conclusion

The selected paper demonstrates a straightforward stacking approach to heart-disease classification and reports 98.53% accuracy. The executed reproduction reached 98.54% accuracy and AUC 1.0000, closely matching the headline stacking accuracy. However, the dataset contains

723 exact duplicate rows and the paper omits several implementation details. Seed sensitivity and a unique-record re-evaluation show that near-perfect performance is not stable evidence of generalisation to unseen patients.

LACSV responds to these limitations by deduplicating before splitting, separating tuning from final testing, calibrating probabilities, using soft voting, and reporting bootstrap uncertainty. On the same leakage-free holdout, both the paper-style stack and LACSV achieved 80.33% accuracy; LACSV had F1 0.8182 versus 0.8125 and AUC 0.8712 versus 0.8734. Because the confidence intervals overlap and the unique dataset is small, the evidence supports methodological improvement rather than a claim of statistically established predictive superiority. External validation on independent clinical data is required before any clinical-use claim.

12. Video Presentation Plan (≤6 minutes)

Time	Segment	What to show/say
0:00-0:40	Problem + selected paper	State the clinical prediction task, paper title, 1,025×14 data, six classifiers and 5-fold stacking.
0:40-1:30	Published method/results	Show Table 11: RF/DT 92.68%, XGB 90.73%, stack 98.53%; explain that some implementation details are missing.
1:30-2:20	Reproduction	Screen-share notebook cells: dataset audit, fixed assumptions, six models + StackingClassifier, generated comparison table.
2:20-3:10	Critical finding	Show duplicate count and explain why identical rows across train/test can inflate generalisation; mention Table 11 metric inconsistency.
3:10-4:30	Your LACSV solution	Show deduplication, untouched holdout, training-only tuning, calibrated LR/SVM/RF soft voting and bootstrap CI.
4:30-5:20	Results	Show leakage-free comparison CSV/ROC. Explain only differences measured under the same protocol.
5:20-6:00	Reflection	Limitations: small unique sample, missing paper details, no external clinical validation; future work: external prospective cohort.

13. Submission Links

Video presentation: https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=d667c1a2-d5a3-4a96-aa57-b4d900bb295e

Complete code, dataset and README files (OneDrive): https://1drv.ms/f/c/bcbdd717d2403d53/IgDx2IZE9HVZRIEG-4CPAf9fAQH4B1FJnweSct2JH71zSNw?e=AXhf5e

14. Generative AI Acknowledgement

GenAI acknowledgement - review and edit this statement so it exactly matches your own use before submission: “Generative AI (ChatGPT) was used for planning, code scaffolding, debugging support,

interpretation of executed outputs, and language/formatting assistance. All experiments were executed on the stated dataset, numerical results were taken from the saved notebook outputs, and the final methodology, analysis, claims and wording were critically reviewed by the student. The student remains responsible for the correctness and academic integrity of the submitted work."

References

[1] M. Bhagat, A. Sharma, and P. Agarwal, "An efficient stacking-based ensemble technique for early heart attack prediction," Multimedia Tools and Applications, vol. 84, no. 30, pp. 36351–36375, 2025, doi: 10.1007/s11042-024-19293-7.

[2] UCI Machine Learning Repository, "Heart Disease," Dataset 45. Accessed: Sep. 25, 2026. https://archive.ics.uci.edu/dataset/45/heart+disease

[3] T. A. Y. Siswa, R. Y. R. Menono, M. W. Hidayatullah, D. I. Rabbil, S. Safitri, and N. D. Yuanti, "Comparison of Machine Learning Classifiers with PSO-Based Optimization and Correlation Feature Selection for Heart Disease Detection," Engineering Proceedings, vol. 137, no. 1, art. 24, 2026, doi: 10.3390/engproc2026137024.

[4] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, art. 100804, 2023, doi: 10.1016/j.patter.2023.100804.

[5] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," BMC Bioinformatics, vol. 7, art. 91, 2006, doi: 10.1186/1471-2105-7-91.

[6] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in Proc. 22nd Int. Conf. Machine Learning (ICML), 2005, pp. 625–632, doi: 10.1145/1102351.1102430.

[7] M. Bhagat, A. Sharma, and P. Agarwal, "Correction: An efficient stacking-based ensemble technique for early heart attack prediction," Multimedia Tools and Applications, vol. 84, no. 30, p. 36377, 2025, doi: 10.1007/s11042-024-20064-7.

[8] Kaggle, "Heart Disease Dataset," johnsmith88/heart-disease-dataset. Accessed: Sep. 25, 2026. https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset.

[9] R. Hidayat, "Dataset Heart Desease," Zenodo, Version 1, Aug. 4, 2024, doi: 10.5281/zenodo.13208473. The distributed heart.csv has MD5 6481ed154d5b8ff83d7c09007aaa6ed0.

Appendix A - Deliverable Checklist

- Completed PDF research report with executed reproduction and proposed-model results.
- Executed Jupyter notebook with all 25 code cells run, outputs visible, and no error outputs.
- Checksum-verified heart.csv included in the final package.
- figures/ and results/ generated by the executed notebook and included.
- README with install, run, and reproduction instructions included.
- Accessible Panopto video link added; video is <=6 minutes with face visible, screen sharing, and clear narration.
- Accessible OneDrive folder link added for the complete code, dataset and README files.
- Review the GenAI acknowledgement so it accurately describes your final use.